Dynamic causal effects I

This version: October 29, 2026

ECMA 31150, Spring 2026

University of Chicago

Department of Economics

Mikkel Plagborg-Moller

Plan for this slide deck

1. Dynamic causal effects
   - Rules vs. shocks
   - Impulse response functions
2. Local projections with observed shock
   - Control variables, lag length selection
   - Standard errors, confidence band
3. Identification from timing restrictions

Causal inference in macroeconomics

- Classic question: how does monetary policy affect the real economy?
- In most countries outside crisis periods, monetary policy operates by manipulating short-term interest rates (U.S.: federal funds rate).
- A simple regression of GDP growth on short-term interest rates does not estimate a causal effect.
  - Simultaneous causality: policy-makers set the interest rate in response to economic developments.
- Even if, hypothetically, monetary policy-makers did not pay attention to the economy, we'd still need to distinguish between two types of interest rate changes with disparate causal effects:
  1. Anticipated changes.
  2. Unanticipated changes: **shocks**.

Causal inference in macroeconomics (cont.)

- Example: suppose policy-makers choose the interest rate according to a **Taylor rule**

$$i_t = \underbrace{c + \phi_i i_{t-1} + \phi_\pi \pi_t + \phi_y y_t}_{\text{systematic } \textbf{policy rule}} + \varepsilon_t.$$

  - i_t: nominal interest rate
  - π_t: inflation
  - y_t: output
  - ε_t: monetary policy shock (idiosyncratic deviation from rule)
- Two-way feedback: $i_t \leftrightarrow (\pi_t, y_t)$.

Causal inference in macroeconomics (cont.)

$$i_t = c + \phi_i i_{t-1} + \phi_\pi \pi_t + \phi_y y_t + \varepsilon_t$$

- Consider a change to the systematic policy rule (e.g., increasing ϕ_i).
  - Economic agents **anticipate** a new equilibrium even in the long run.
  - Agents fundamentally change their behavior (e.g., their formula for forecasting the future).
- Now consider a one-off surprise change to ε_t (a shock).
  - Agents update expectations about short-run economic conditions, but the structure of the economy is unchanged.
- The causal effects of these two types of changes are not the same.

Causal inference in macroeconomics (cont.)

- In practice, we do not cleanly observe many structural changes in policy rules in historical data. Difficult (but not impossible) to estimate causal effects of changing the policy *rule*.
- Instead we will focus on estimating causal effects of *shocks*: deviations from the policy rule.
- Where do policy shocks come from?
  - Idiosyncratic preferences of rotating cast of policy-makers (e.g., inflation hawks/doves).
  - One-off political events.
  - Private information of policy-makers.
  - Desire to surprise markets (but can't do this all the time).

You can fool some people sometimes, but you can't fool all the people all the time. — Bob Marley

Structural vector moving average model

- Let y_t be an observed macro time series.
  - GDP, inflation, interest rate, etc.
- Let ε_t be an $n_\varepsilon \times 1$ vector of potentially unobserved shocks.
- **Structural vector moving average** (SVMA) model:

$$y_t = \mu + \mathbf{\Theta}_0 \varepsilon_t + \mathbf{\Theta}_1 \varepsilon_{t-1} + \mathbf{\Theta}_2 \varepsilon_{t-2} + \cdots = \mu + \mathbf{\Theta}(L)\varepsilon_t,$$

where $\mathbf{\Theta}(L) = \sum_{\ell=0}^{\infty} \mathbf{\Theta}_\ell L^\ell$ is of dimension $1 \times n_\varepsilon$.

Structural vector moving average model (cont.)

- The observed variable can depend on all lags of all shocks:

$$y_t = \mu + \sum_{j=1}^{n_\varepsilon} \sum_{\ell=0}^{\infty} \Theta_{\ell j} \varepsilon_{j,t-\ell}.$$

  - $\Theta_{\ell j}$: j-th element of $\mathbf{\Theta}_\ell$.

- Apart from linearity, SVMA model places *no restrictions* on the **propagation mechanism** from shocks to observed macro outcomes.

- Encompasses any linear theoretical macro model you can think of.

  - Exercise: if $y_t = c + \rho y_{t-1} + \varepsilon_t$ with $|\rho| < 1$, how can you represent y_t in SVMA form?

Structural vector moving average model: shocks

$$y_t = \mu + \boldsymbol{\Theta}_0 \varepsilon_t + \boldsymbol{\Theta}_1 \varepsilon_{t-1} + \boldsymbol{\Theta}_2 \varepsilon_{t-2} + \cdots = \mu + \boldsymbol{\Theta}(L)\varepsilon_t$$

- Consistent with our earlier discussion, we assume that shocks are serially uncorrelated: $\varepsilon_t \sim WN(\boldsymbol{D})$.
  - A "shock" has to be unpredictable from past information.
- Moreover, we assume that the variance-covariance matrix $\boldsymbol{D}$ is diagonal, so shocks are **mutually uncorrelated**:

$$\boldsymbol{D} = \text{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_{n_\varepsilon}^2) \quad \Longrightarrow \quad E[\varepsilon_{it}\varepsilon_{jt}] = 0 \quad \text{for } i \neq j.$$

  - Different shocks correspond to conceptually distinct external forces that drive the economy (monetary policy, technology, climate, etc.).
  - If two supposed "shocks" were correlated, they should be viewed as both being caused by a third, more fundamental shock.

Structural vector moving average model (cont.)

$$y_t = \mu + \boldsymbol{\Theta}(L)\boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim WN(\boldsymbol{D}), \quad \boldsymbol{D} = \text{diag}(\sigma_1^2, \sigma_2^2, \ldots, \sigma_{n_\varepsilon}^2)$$

- SVMA model casts the Frisch-Slutsky-Yule **impulse-propagation framework** as a causal model (recall our first lecture).
- Meaningful to think of manipulating one shock while holding all other shocks constant.
- Presumably, there are in reality thousands of different kinds of shocks that drive the economy (not just policy, but also surprise changes to economic fundamentals such as technology).

SVMA: impulse response function

- How does a shock, say the first one ε_{1t}, causally affect the outcomes now and in the future: y_t, y_{t+1}, y_{t+2}, ...?
- Let $h \geq 0$. Let's express y_{t+h} as a function of ε_{1t} plus other stuff:

$$\begin{aligned} y_{t+h} &= \mu + \sum_{j=1}^{n_\varepsilon} \sum_{\ell=0}^{\infty} \Theta_{\ell j} \varepsilon_{j,t+h-\ell} \\ &= \mu + \Theta_{01} \varepsilon_{1,t+h} + \Theta_{11} \varepsilon_{1,t+h-1} + \Theta_{21} \varepsilon_{1,t+h-2} + \cdots \\ &\quad + \sum_{j=2}^{n_\varepsilon} \sum_{\ell=0}^{\infty} \Theta_{\ell j} \varepsilon_{j,t+h-\ell} \\ &= \mu + \Theta_{h1} \varepsilon_{1t} + \sum_{\ell \neq h} \Theta_{\ell 1} \varepsilon_{1,t+h-\ell} + \sum_{j=2}^{n_\varepsilon} \sum_{\ell=0}^{\infty} \Theta_{\ell j} \varepsilon_{j,t+h-\ell}. \end{aligned}$$

- Causal effect on y_{t+h} of increasing ε_{1t} by 1 unit: $\boxed{\Theta_{h1}}$.

SVMA: impulse response function (cont.)

- If ε_{1t} is increased by 1 unit, holding everything else constant (including past and future shocks)...
  - Current period (**on impact**): y_t changes by Θ_{01}.
  - Next period: y_{t+1} changes by Θ_{11}.
  - Two periods in the future: y_{t+2} changes by Θ_{21}...
- The MA parameter Θ_{h1} is called the **impulse response** at **horizon** h.
- We call $(\Theta_{01}, \Theta_{11}, \Theta_{21}, \dots)$ the **impulse response function** (IRF) – or **dynamic causal effects** – of $\{y_t\}$ with respect to the shock $\{\varepsilon_{1t}\}$.

Impulse response function: examples

- SVMA(2) model:

$$y_t = \varepsilon_{1t} - 0.5\varepsilon_{1,t-1} + \varepsilon_{2,t-1} + 0.2\varepsilon_{1,t-2}$$

  - IRF of y_t wrt. ε_{1t}: $(1, -0.5, 0.2, 0, 0, \dots)$.
  - IRF of y_t wrt. ε_{2t}: $(0, 1, 0, 0, 0, \dots)$.

- Exercise: show that in the AR(1) model $y_t = c + \rho y_{t-1} + \varepsilon_t$, the IRF of y_t wrt. ε_t is $(1, \rho, \rho^2, \rho^3, \dots)$.

Impulse response function: estimation

- Suppose we directly measure the shock ε_{1t}.
- From our earlier derivation:

$$y_{t+h} = \mu + \Theta_{h1}\varepsilon_{1t} + \underbrace{\sum_{\ell \neq h} \Theta_{\ell 1}\varepsilon_{1,t+h-\ell} + \sum_{j=2}^{n_\varepsilon}\sum_{\ell=0}^{\infty} \Theta_{\ell j}\varepsilon_{j,t+h-\ell}}_{\equiv v_{h,t+h}}.$$

- By our assumptions on the shocks, ε_{1t} is uncorrelated with each of the terms that appear in $v_{h,t+h}$:

$$E[\varepsilon_{1t}\varepsilon_{j,t+h-\ell}] = 0 \quad \text{if either } j \neq 1 \text{ or } \ell \neq h.$$

- We can therefore consistently estimate the impulse response Θ_{h1} through an OLS regression of y_{t+h} on ε_{1t}.
- Notation: to emphasize that ε_{1t} is observed, I will call it x_t.

Local projections

- Separately for each horizon $h = 0, 1, \ldots, h_{\max}$ of interest, we run an OLS regression

$$y_{t+h} = \hat{c}_h + \hat{\beta}_h x_t + \hat{v}_{h,t+h}.$$

- Then $\hat{\beta}_h$ consistently estimates the impulse response Θ_{h1}. The estimated IRF up to horizon $h_{\max}$ is $(\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_{h_{\max}})$.
- These regressions are called **local projections** (LPs).
- We could do this for different macro observables y_t.
- The true regression error term $v_{h,t+h}$ is serially correlated, which suggests reporting Newey-West SE (more on this later).

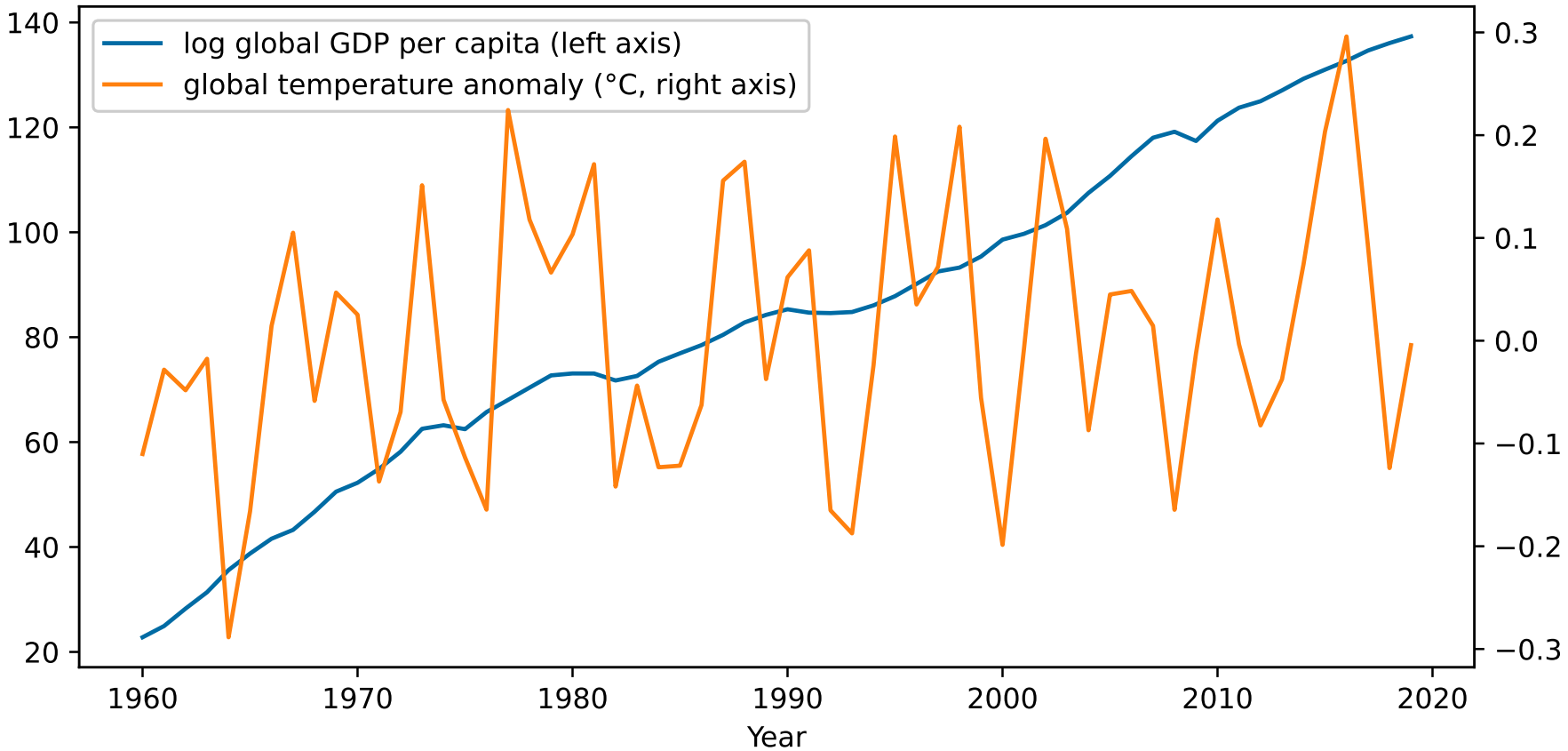
THE
QUARTERLY JOURNAL
OF ECONOMICS

Vol. 141 2026 Issue 2

THE MACROECONOMIC IMPACT OF CLIMATE CHANGE: GLOBAL VERSUS LOCAL TEMPERATURE*

ADRIEN BILAL AND DIEGO R. KÄNZIG

This article estimates that the macroeconomic damages from climate change are an order of magnitude larger than previously thought. Exploiting natural global temperature variability, we find that 1°C warming reduces world GDP by over 20% in the long run. Global temperature correlates strongly with extreme climatic events, unlike country-level temperature used in previous work, explaining our larger estimate. We use this evidence to estimate damage functions in a neoclassical growth model. Business-as-usual warming implies a present welfare loss of more than 30%, and a social cost of carbon in excess of $1,200 per ton. These impacts suggest that unilateral decarbonization policy is cost-effective for large countries such as the United States. *JEL codes:* E01, E23, F18, O44, Q54, Q56.



Temperature anomaly: deviation of global temperature from medium-run trend

```
max_horizon = 10
T = df[['l_gdp', 'temp_anom']].dropna().shape[0]
nw_bw = int(np.ceil(0.75 * T**(1/3)))

irf = np.empty(max_horizon + 1)
se  = np.empty(max_horizon + 1)
for h in range(max_horizon + 1):
    y = (df['l_gdp'].shift(-h) - df['l_gdp'].shift(1)).rename(f'dl_gdp_h{h}')
    X = sm.add_constant(df[['temp_anom']])
    reg = sm.OLS(y, X, missing='drop').fit(
        cov_type='HAC', cov_kwds={'maxlags': nw_bw - 1}
        )
    irf[h] = reg.params['temp_anom']
    se[h]  = reg.bse['temp_anom']
```

- Note: since GDP has a clear upward trend over time, I transform the dependent variable in the regression to "long differences" $y_{t+h} - y_{t-1}$, which look more stationary.
- The long differences $(y_t - y_{t-1}), (y_{t+1} - y_{t-1}), (y_{t+2} - y_{t-1}), \ldots$ are labeled `dl_gdp_h0`, `dl_gdp_h1`, `dl_gdp_h2`, ...

$h = 0$

OLS Regression Results

Dep. Variable:	dl_gdp_h0	R-squared:	0.000
Model:	OLS	Adj. R-squared:	-0.017
Method:	Least Squares	F-statistic:	0.004007
Date:	Wed, 22 Apr 2026	Prob (F-statistic):	0.950
Time:	13:19:42	Log-Likelihood:	-100.84
No. Observations:	59	AIC:	205.7
Df Residuals:	57	BIC:	209.8
Df Model:	1		
Covariance Type:	HAC		

	coef	std err	z	P>|z|	[0.025	0.975]
const	1.9419	0.217	8.936	0.000	1.516	2.368
temp_anom	-0.0936	1.479	-0.063	0.950	-2.992	2.805

$\hat{\beta}_0$

Omnibus:	3.905	Durbin-Watson:	1.268
Prob(Omnibus):	0.142	Jarque-Bera (JB):	3.212
Skew:	-0.565	Prob(JB):	0.201
Kurtosis:	3.176	Cond. No.	8.02

Notes:
[1] Standard Errors are heteroscedasticity and autocorrelation robust (HAC) using 2 lags and without small sample correction

$h = 1$

OLS Regression Results

Dep. Variable:	dl_gdp_h1	R-squared:	0.002
Model:	OLS	Adj. R-squared:	-0.016
Method:	Least Squares	F-statistic:	0.08369
Date:	Wed, 22 Apr 2026	Prob (F-statistic):	0.773
Time:	13:19:42	Log-Likelihood:	-128.62
No. Observations:	58	AIC:	261.2
Df Residuals:	56	BIC:	265.4
Df Model:	1		
Covariance Type:	HAC		

	coef	std err	z	P>|z|	[0.025	0.975]
const	3.8956	0.414	9.418	0.000	3.085	4.706
temp_anom	-0.7030	2.430	-0.289	0.772	-5.465	4.059

$\hat{\beta}_1$

Omnibus:	2.629	Durbin-Watson:	0.721
Prob(Omnibus):	0.269	Jarque-Bera (JB):	2.500
Skew:	-0.491	Prob(JB):	0.286
Kurtosis:	2.732	Cond. No.	7.96

Notes:
[1] Standard Errors are heteroscedasticity and autocorrelation robust (HAC) using 2 lags and without small sample correction

$h = 2$

OLS Regression Results

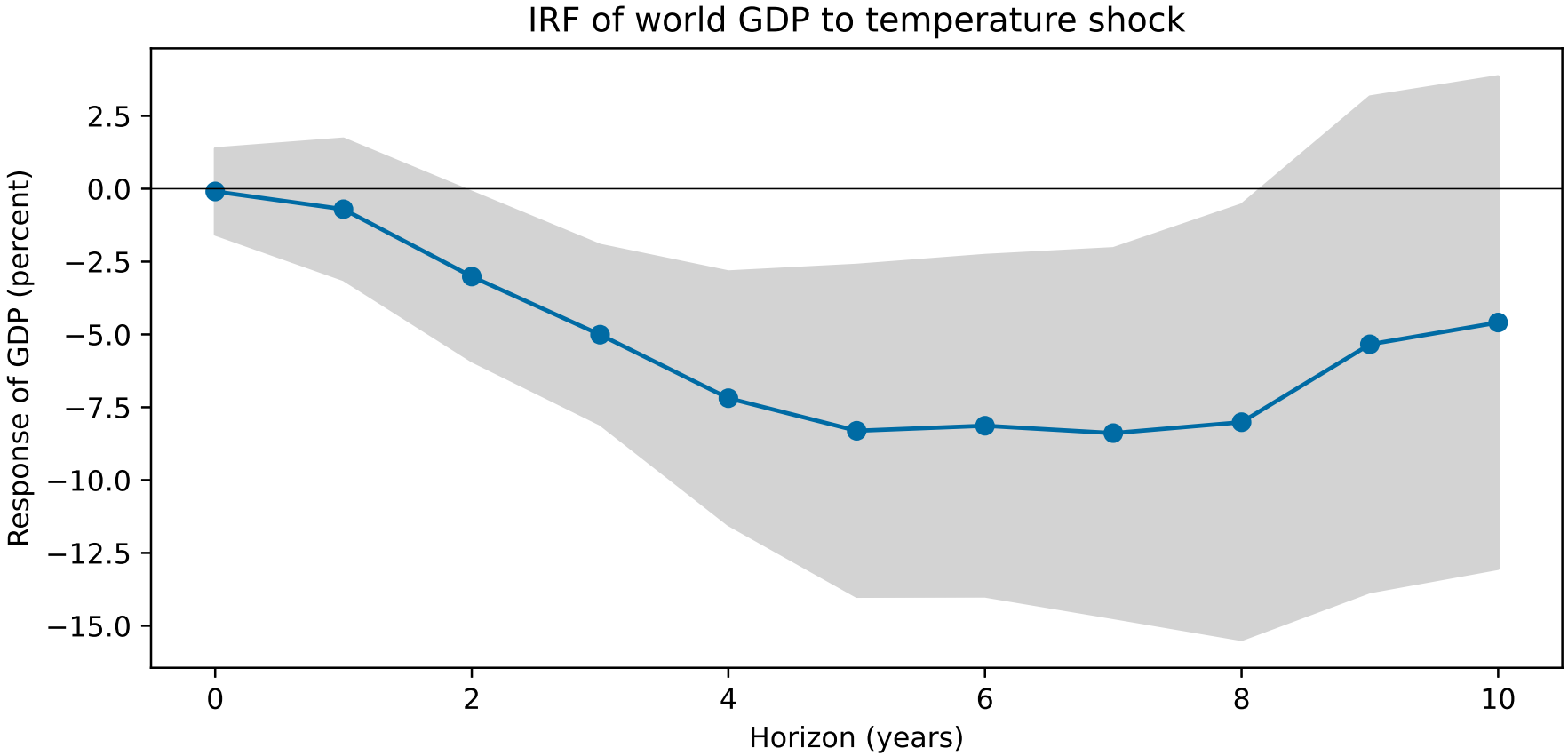
Dep. Variable:	dl_gdp_h2	R-squared:	0.017
Model:	OLS	Adj. R-squared:	-0.000
Method:	Least Squares	F-statistic:	1.068
Date:	Wed, 22 Apr 2026	Prob (F-statistic):	0.306
Time:	13:19:42	Log-Likelihood:	-140.40
No. Observations:	57	AIC:	284.8
Df Residuals:	55	BIC:	288.9
Df Model:	1		
Covariance Type:	HAC		

	coef	std err	z	P>|z|	[0.025	0.975]
const	5.8527	0.577	10.140	0.000	4.721	6.984
temp_anom	-3.0120	2.914	-1.034	0.301	-8.723	2.699

$\hat{\beta}_2$

Omnibus:	1.677	Durbin-Watson:	0.418
Prob(Omnibus):	0.432	Jarque-Bera (JB):	1.664
Skew:	-0.375	Prob(JB):	0.435
Kurtosis:	2.628	Cond. No.	7.96

Notes:
[1] Standard Errors are heteroscedasticity and autocorrelation robust (HAC) using 2 lags and without small sample correction



Blue dots: estimated IRF

Grey region: 68% **confidence band** formed by the collection of 68% confidence intervals for each of the impulse responses

Control variables

- The LP estimator is consistent under our assumptions. But we can obtain more *efficient* estimates by controlling for lagged variables in our regressions:

$$y_{t+h} = \hat{c}_h + \hat{\beta}_h x_t + \hat{\boldsymbol{\delta}}_1' \boldsymbol{w}_{t-1} + \cdots + \hat{\boldsymbol{\delta}}_p' \boldsymbol{w}_{t-p} + \hat{v}_{h,t+h},$$

  where $\boldsymbol{w}_t$ is a vector of data series (including y_t, x_t, and potentially other variables).

- If $x_t = \varepsilon_{1t}$ is a shock, it's uncorrelated with $\boldsymbol{w}_{t-1}, \ldots, \boldsymbol{w}_{t-p}$, so the controls are not needed to remove omitted variable bias.

- But the controls "soak up" part of the variance of the regression residual, which leads to smaller SE.

Control variables and SE

$$y_{t+h} = \hat{c}_h + \hat{\beta}_h x_t + \hat{\boldsymbol{\delta}}_1' \boldsymbol{w}_{t-1} + \cdots + \hat{\boldsymbol{\delta}}_p' \boldsymbol{w}_{t-p} + \hat{v}_{h,t+h}$$

- In fact, it can be shown that, under some reasonable assumptions, if we control for enough variables in $\boldsymbol{w}_t$ and enough lags p, then we can use heteroskedasticity-robust SE for $\hat{\beta}_h$ rather than Newey-West.
  - Simulations show that the former SE typically work better in finite samples. Moreover, they do not require choosing a bandwidth.
  - Though $\{v_{h,t+h}\}$ is serially correlated, the product $\{x_t \times v_{h,t+h}\}$ is typically much less so if x_t is a shock. Recall that it's the LRV of the product process that matters for SE.
- Should control for variables that are strong predictors of y_t.
- Can select p using AIC applied to a VAR in all the variables in the regression.

```
controls = ['l_gdp', 'temp_anom', 'lnpoil_wti',
            'treasury1y', 'recessiondates']
var_model = sm.tsa.VAR(df[controls].dropna())
lag_order = var_model.select_order(maxlags=4)
print(lag_order.summary())
```

Results:

```
VAR Order Selection (* highlights the minimums)
=================================================
      AIC         BIC         FPE         HQIC
-------------------------------------------------
0       9.783       9.964   1.773e+04       9.853
1     0.5904*      1.675*      1.812*      1.011*
2      0.6255       2.615       1.918       1.397
3      0.8745       3.768       2.602       1.996
4      0.9227       4.720       3.050       2.395
-------------------------------------------------
```

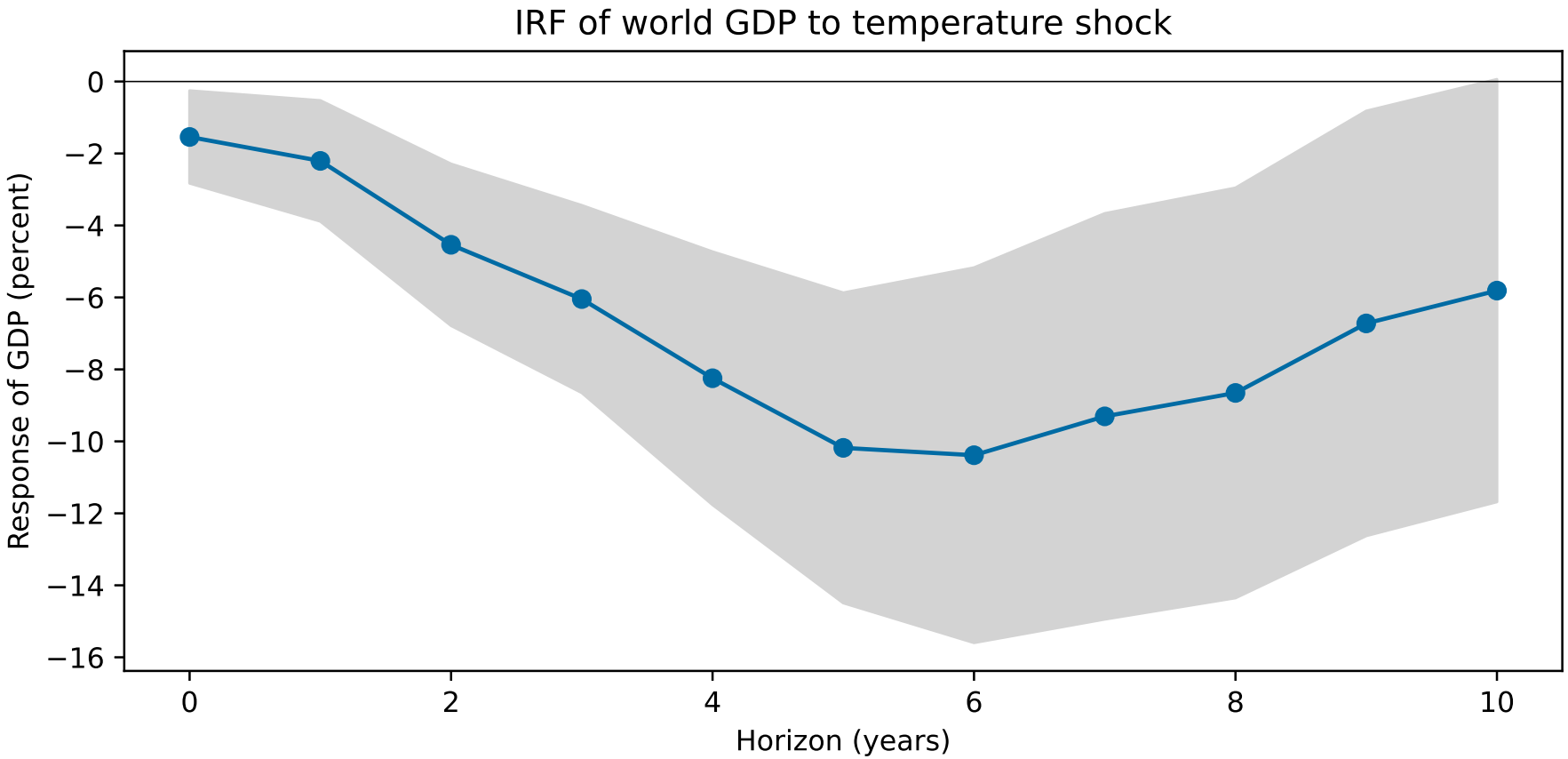
I'll use $p = 2$ to be on the safe side.

```
def local_projection(df, outcome, shock, controls, max_horz):
    """Local projection IRF with SE"""
    irf = np.empty(max_horz + 1)
    se  = np.empty(max_horz + 1)
    for h in range(max_horz + 1):
        y = df[outcome].shift(-h)
        X = sm.add_constant(df[[shock] + controls])
        reg = sm.OLS(y, X, missing='drop').fit(cov_type='HC1')
        irf[h] = reg.params[shock]
        se[h]  = reg.bse[shock]
    return irf, se
```

```
# Create lagged data
nlag = 2
df = df.assign(**{
    f'{var}_lag{l}': df[var].shift(l)
    for var in controls for l in range(1, nlag + 1)
})
lag_cols = [f'{var}_lag{l}' for var in controls for l in range(1, nlag + 1)]

# LP with lagged controls
irf, se = local_projection(df, 'l_gdp', 'temp_anom', lag_cols, max_horizon)
```

Note: the outcome can now just be y_{t+h} rather than $(y_{t+h} - y_{t-1})$, since the long differencing is absorbed by the lagged control y_{t-1}. Effectively, this lagged control removes any trend in the outcome.



Controls: 2 lags of outcome, shock, oil price, 1-year Treasury rate, global recession indicator

The confidence band is narrower than w/o controls – increased efficiency!

Is the "shock" a shock?

- x_t is supposed to be a shock, and thus unpredictable.
- We can test this by regressing this supposed shock on lagged data (including lags of itself).
- In the application, we should be worried: the IRF estimates changed quite a bit when we added lagged controls, even though x_t ought to be uncorrelated with those lags.

OLS Regression Results

Dep. Variable:	temp_anom	R-squared:	0.155
Model:	OLS	Adj. R-squared:	0.074
Method:	Least Squares	F-statistic:	2.486
Date:	Tue, 21 Apr 2026	Prob (F-statistic):	0.0430
Time:	09:23:44	Log-Likelihood:	42.917
No. Observations:	58	AIC:	-73.83
Df Residuals:	52	BIC:	-61.47
Df Model:	5		
Covariance Type:	HC1		

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.0483	0.042	1.144	0.253	-0.034	0.131
dl_gdp_lag1	-0.0062	0.012	-0.507	0.612	-0.030	0.018
dlnpoil_lag1	-0.0004	0.001	-0.572	0.567	-0.002	0.001
temp_anom_lag1	0.2340	0.116	2.023	0.043	0.007	0.461
treasury1y_lag1	-0.0014	0.005	-0.293	0.769	-0.011	0.008
recessiondates_lag1	-0.0778	0.036	-2.166	0.030	-0.148	-0.007

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

`dl_gdp`: $\Delta \log(\text{GDP}_t)$, `dlnpoil`: $\Delta \log(\text{oilprice}_t)$, `treasury1y`: 1-year Treasury interest rate, `recessiondates`: indicator for global recession

OLS Regression Results

Dep. Variable:	temp_anom	R-squared:	0.155
Model:	OLS	Adj. R-squared:	0.074
Method:	Least Squares	F-statistic:	2.486
Date:	Tue, 21 Apr 2026	Prob (F-statistic):	0.0430
Time:	09:23:44	Log-Likelihood:	42.917
No. Observations:	58	AIC:	-73.83
Df Residuals:	52	BIC:	-61.47
Df Model:	5		
Covariance Type:	HC1		

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.0483	0.042	1.144	0.253	-0.034	0.131
dl_gdp_lag1	-0.0062	0.012	-0.507	0.612	-0.030	0.018
dlnpoil_lag1	-0.0004	0.001	-0.572	0.567	-0.002	0.001
temp_anom_lag1	0.2340	0.116	2.023	0.043	0.007	0.461
treasury1y_lag1	-0.0014	0.005	-0.293	0.769	-0.011	0.008
recessiondates_lag1	-0.0778	0.036	-2.166	0.030	-0.148	-0.007

Notes:
[1] Standard Errors are heteroscedasticity robust (HC1)

Significant predictability

`dl_gdp`: $\Delta \log(\text{GDP}_t)$, `dlnpoil`: $\Delta \log(\text{oilprice}_t)$, `treasury1y`: 1-year Treasury interest rate, `recessiondates`: indicator for global recession

Residualized regression

- Oops – our "shock" is predictable! But all is not lost.
- By the **residual regression** (Frisch-Waugh-Lovell) theorem, the $\hat{\beta}_h$ from the regression with controls,

$$y_{t+h} = \hat{c}_h + \hat{\beta}_h x_t + \hat{\boldsymbol{\delta}}_1' \boldsymbol{w}_{t-1} + \cdots + \hat{\boldsymbol{\delta}}_p' \boldsymbol{w}_{t-p} + \hat{v}_{h,t+h},$$

  can equivalently be obtained from a bivariate regression

$$\boxed{y_{t+h} = \hat{\beta}_h \tilde{x}_t + \tilde{v}_{h,t+h}},$$

  where $\tilde{x}_t \equiv$ residual from a first-step regression of x_t on $\boldsymbol{w}_{t-1}, \ldots, \boldsymbol{w}_{t-p}$ (and an intercept).
- So our LP with controls is really measuring the impulse responses wrt. the **residualized shock** $\tilde{x}_t$.
  - This is the one-step-ahead *forecast error* of the temperature anomaly.

Residualized regression

- Even if x_t is predictable, $\tilde{x}_t$ may still be measuring a meaningful shock.
- For example, suppose the temperature anomaly follows an ADL model:

$$\texttt{temp_anom}_t = \rho \times \texttt{temp_anom}_{t-1} + b \times \texttt{recessiondates}_{t-1} + \varepsilon_{1t},$$

  where ε_{1t} is a truly unpredictable climate shock.
- Then $\tilde{x}_t \approx \varepsilon_{1t}$ (with equality as $T \to \infty$).

```
# Multivariate regression
X = sm.add_constant(df[['temp_anom'] + lag_cols])
print(sm.OLS(df['l_gdp'], X, missing='drop').fit().summary())

# Residualized regression
df['temp_anom_resid'] = sm.OLS(df['temp_anom'],
    sm.add_constant(df[lag_cols]), missing='drop').fit().resid
print(sm.OLS(df['l_gdp'],
             sm.add_constant(df['temp_anom_resid']),
             missing='drop').fit().summary())
```

OLS Regression Results

Dep. Variable:	l_gdp	R-squared:	0.999
Model:	OLS	Adj. R-squared:	0.999
Method:	Least Squares	F-statistic:	3946.
Date:	Wed, 22 Apr 2026	Prob (F-statistic):	1.56e-64
Time:	13:57:17	Log-Likelihood:	-79.599
No. Observations:	58	AIC:	183.2
Df Residuals:	46	BIC:	207.9
Df Model:	11		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	4.5430	1.011	4.494	0.000	2.508	6.578
temp_anom	-1.5414	1.402	-1.099	0.277	-4.364	1.281
l_gdp_lag1	1.0338	0.165	6.248	0.000	0.701	1.367
l_gdp_lag2	-0.0646	0.162	-0.399	0.692	-0.390	0.261
temp_anom_lag1	0.3568	1.389	0.257	0.798	-2.438	3.152
temp_anom_lag2	-0.3467	1.489	-0.233	0.817	-3.345	2.651
lnpoil_wti_lag1	-0.0036	0.008	-0.469	0.641	-0.019	0.012
lnpoil_wti_lag2	0.0079	0.007	1.135	0.262	-0.006	0.022
treasury1y_lag1	-0.1830	0.135	-1.359	0.181	-0.454	0.088
treasury1y_lag2	-0.0046	0.140	-0.033	0.974	-0.287	0.278
recessiondates_lag1	-1.2033	0.442	-2.725	0.009	-2.092	-0.315
recessiondates_lag2	-0.1892	0.486	-0.389	0.699	-1.168	0.790

OLS Regression Results

Dep. Variable:	l_gdp	R-squared:	0.000
Model:	OLS	Adj. R-squared:	-0.018
Method:	Least Squares	F-statistic:	0.001557
Date:	Wed, 22 Apr 2026	Prob (F-statistic):	0.969
Time:	13:57:17	Log-Likelihood:	-278.27
No. Observations:	58	AIC:	560.5
Df Residuals:	56	BIC:	564.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	86.0141	3.920	21.940	0.000	78.161	93.868
temp_anom_resid	-1.5414	39.064	-0.039	0.969	-79.796	76.713

Note: Don't use the SE from the residualized regression!

Contemporaneous control variables

- We may still worry that the one-step-ahead forecast error of the temperature anomaly is contaminated by economic shocks.
  - Shock to economic activity $\Longrightarrow$ higher CO_2 emissions $\Longrightarrow$ spike in temperature.
  - If this feedback happens fast (within the year), then

$$\tilde{x}_t \approx \text{climate shock} + \text{economic activity shock}.$$

  - So our LP would not be estimating the effect of a pure climate shock.
- We can try to address this by adding *contemporaneous* controls $\boldsymbol{r}_t$:

$$y_{t+h} = \hat{c}_h + \hat{\beta}_h x_t + \hat{\boldsymbol{\delta}}_0' \boldsymbol{r}_t + \hat{\boldsymbol{\delta}}_1' \boldsymbol{w}_{t-1} + \cdots + \hat{\boldsymbol{\delta}}_p' \boldsymbol{w}_{t-p} + \hat{v}_{h,t+h}.$$

- Implicit residualized shock $\tilde{x}_t \equiv$ residual from regressing x_t on $\boldsymbol{r}_t, \boldsymbol{w}_{t-1}, \ldots, \boldsymbol{w}_{t-p}$.

Contemporaneous control variables (cont.)

- Suppose x_t equals the shock of interest plus a contamination ζ_t:

$$x_t = \varepsilon_{1t} + \zeta_t, \quad \text{where} \quad \text{Cov}(\varepsilon_{1t}, \zeta_t) = 0.$$

- ζ_t could depend on other shocks $\varepsilon_{2t}, \ldots, \varepsilon_{n_\varepsilon t}$ as well as lagged variables.
- Then the LP's residualized shock $\tilde{x}_t$ will equal ε_{1t} (asymptotically) provided that *both* the following conditions hold:
  1. $\text{Cov}(\boldsymbol{r}_t, \varepsilon_{1t}) = \mathbf{0}$,
  2. ζ_t is perfectly (linearly) explained by $\boldsymbol{r}_t, \boldsymbol{w}_{t-1}, \ldots, \boldsymbol{w}_{t-p}$.
- Exercise: why is this true?

Identification from timing restrictions

$$x_t = \varepsilon_{1t} + \zeta_t, \quad \text{where} \quad \text{Cov}(\varepsilon_{1t}, \zeta_t) = 0$$

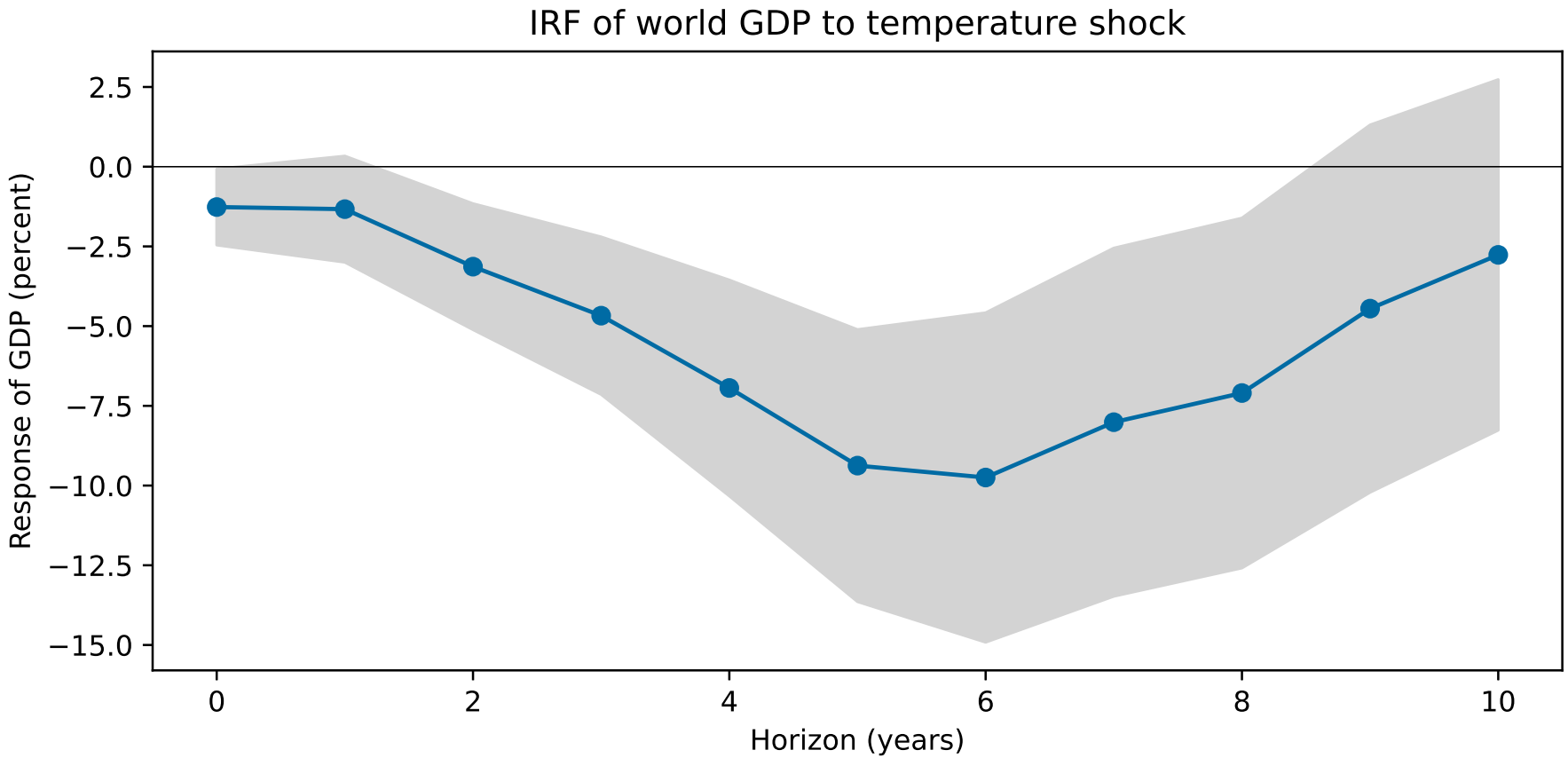
1. $\text{Cov}(\boldsymbol{r}_t, \varepsilon_{1t}) = \mathbf{0}$.
   - Interpretation: the contemporaneous controls do not respond to the shock of interest ε_{1t} on impact (i.e., within the period).
2. ζ_t is perfectly (linearly) explained by $\boldsymbol{r}_t, \boldsymbol{w}_{t-1}, \ldots, \boldsymbol{w}_{t-p}$.
   - Interpretation: the contaminating shocks in ζ_t must be the main shocks that affect $\boldsymbol{r}_t$ on impact.

- This is called **identification from timing restrictions**.
  - We identify the causal effect of $\tilde{x}_t = \varepsilon_{1t}$, even though the shock is not directly observed.
  - But this requires making assumptions on the *timing* of when various shocks affect the variables in $\boldsymbol{r}_t$.

Identification from timing restrictions: example

- The timing assumptions must be defended based on economic theory/intuition. Need to be explicit about what kinds of contaminating shocks we're worried about.
- Bilal & Känzig (2026) set $\boldsymbol{r}_t = \texttt{recessiondates}_t$ in order to control for contamination from economic shocks.
- Implicit timing assumptions:
  1. Global recessions can only respond to the climate shock ε_{1t} with a lag (not within the year).
  2. The economic shock that potentially affects the temperature anomaly is the main driver of global recessions in the short run.

```
# LP with contemporaneous control
control_cols = ['recessiondates'] + lag_cols
irf, se = local_projection(df, 'l_gdp', 'temp_anom', control_cols, max_horizon)
```



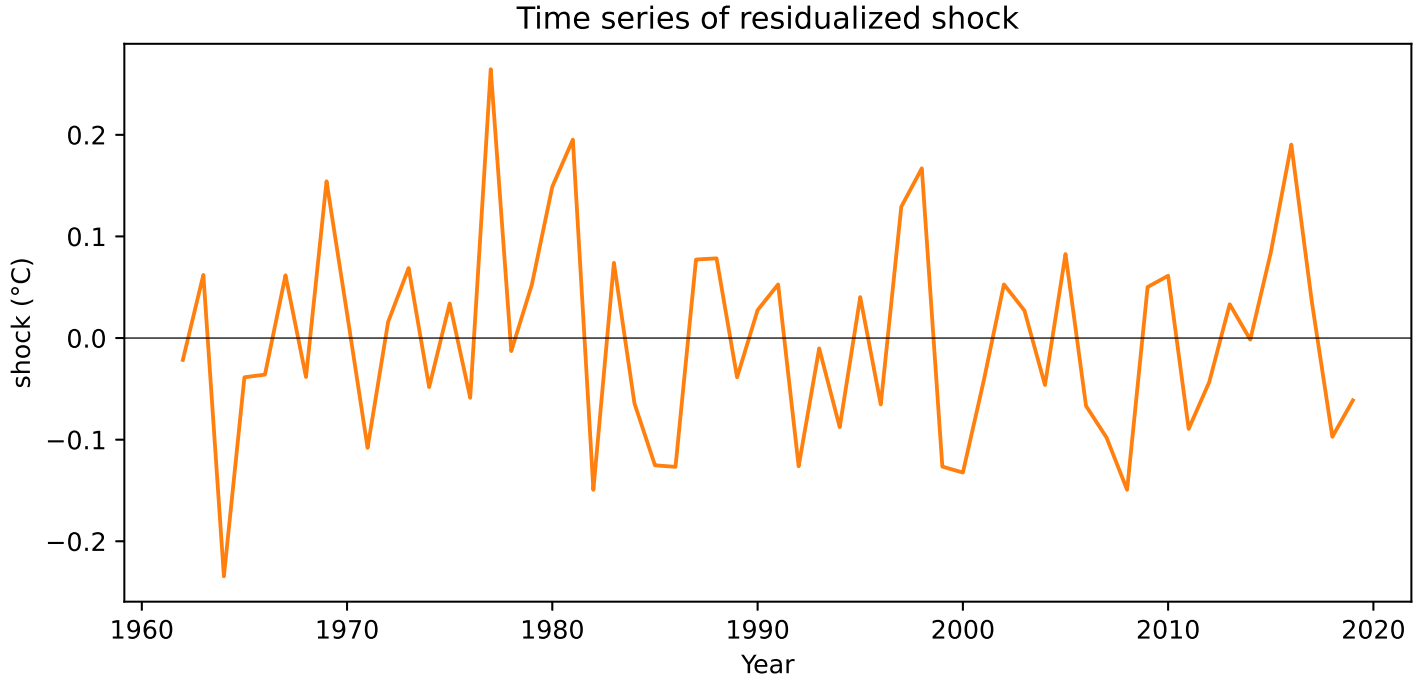
```
# Estimate shock on full sample, using all controls
df['temp_anom_resid'] = sm.OLS(df['temp_anom'],
                               sm.add_constant(df[control_cols]),
                               missing='drop').fit().resid

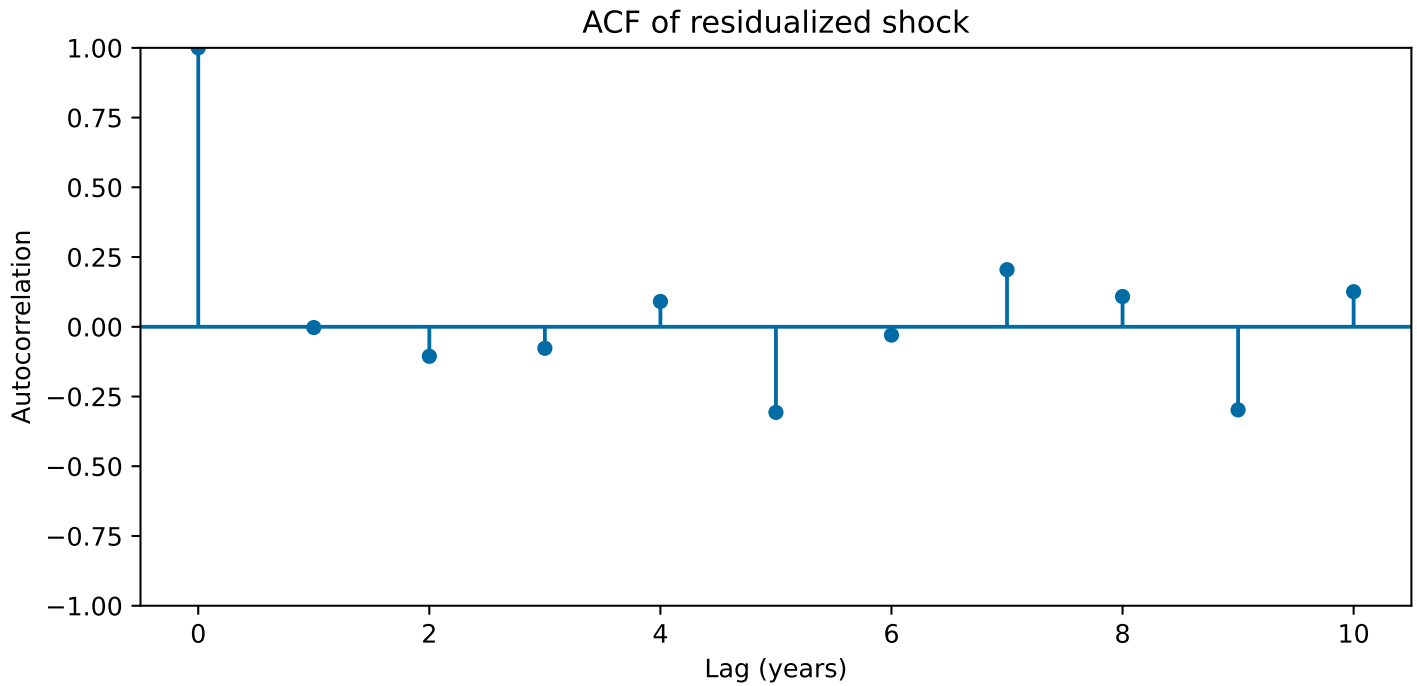
# Summary statistics
print(df['temp_anom_resid'].describe())
```

Results:

```
count    5.800000e+01
mean    -6.849406e-15
std      9.938751e-02
min     -2.344768e-01
25%     -6.508441e-02
50%     -5.863778e-03
75%      6.166066e-02
max      2.645975e-01
Name: temp_anom_resid, dtype: float64
```

Note: a 1 unit (i.e., 1°C) climate shock is 4 $\times$ largest observed shock. Helpful for interpreting magnitudes in the IRF plots.


Time series of residualized shock
shock (°C)
0.2
0.1
0.0
−0.1
−0.2
1960
1970
1980
1990
2000
2010
2020
Year


ACF of residualized shock
Autocorrelation
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
2
4
6
8
10
Lag (years)

Summary: LP with controls

- Control for variables that are strong predictors of *either* the outcome y_t *or* the impulse variable x_t.
  - At a minimum, control for lags of y_t and x_t.
- Use economic theory or prior experience as a guide to select variables.
- Use AIC to select lag length p.
- Important to use lagged controls even if we directly observe a shock: increases efficiency and allows us to avoid Newey-West SE.
- Use contemporaneous controls to remove confounding from other shocks. But must defend the timing assumptions!
- Given controls, report diagnostics for residualized shock $\tilde{x}_t$.

Importance of theory

- LPs are easy to run but not always easy to interpret.
- The residualized shock $\tilde{x}_t$ must correspond to something specific in relevant theoretical macro models.
- Examples:
  - ✓ One could write a model of how climate interacts with the economy. One could imagine an external "climate shock" in such a model.
  - ✗ Suppose we ran an LP on $x_t = \text{GDP}_t$. What is this supposed "GDP shock"? In most macro models, GDP is driven by multiple shocks: technology, monetary policy, "animal spirits", etc.
- Nobody will be impressed by your LP results if you don't have a concrete justification/interpretation for your identified shock.

What did we learn?

- SVMA model as a dynamic causal framework
  - Changes to policy rules vs. shocks
  - IRF
- LP estimation of IRFs
  - Controls, lag length, SE
  - Residualized shock, shock diagnostics
- Identification from timing restrictions

Looking ahead

- How important is our shock $\tilde{x}_t$ for the dynamics of y_t, relative to everything else that drives y_t?
- What if $\tilde{x}_t$ is contaminated with measurement error?
- Small-sample bias of LP and how to deal with it
- How can we exploit panel data to estimate dynamic causal effects?